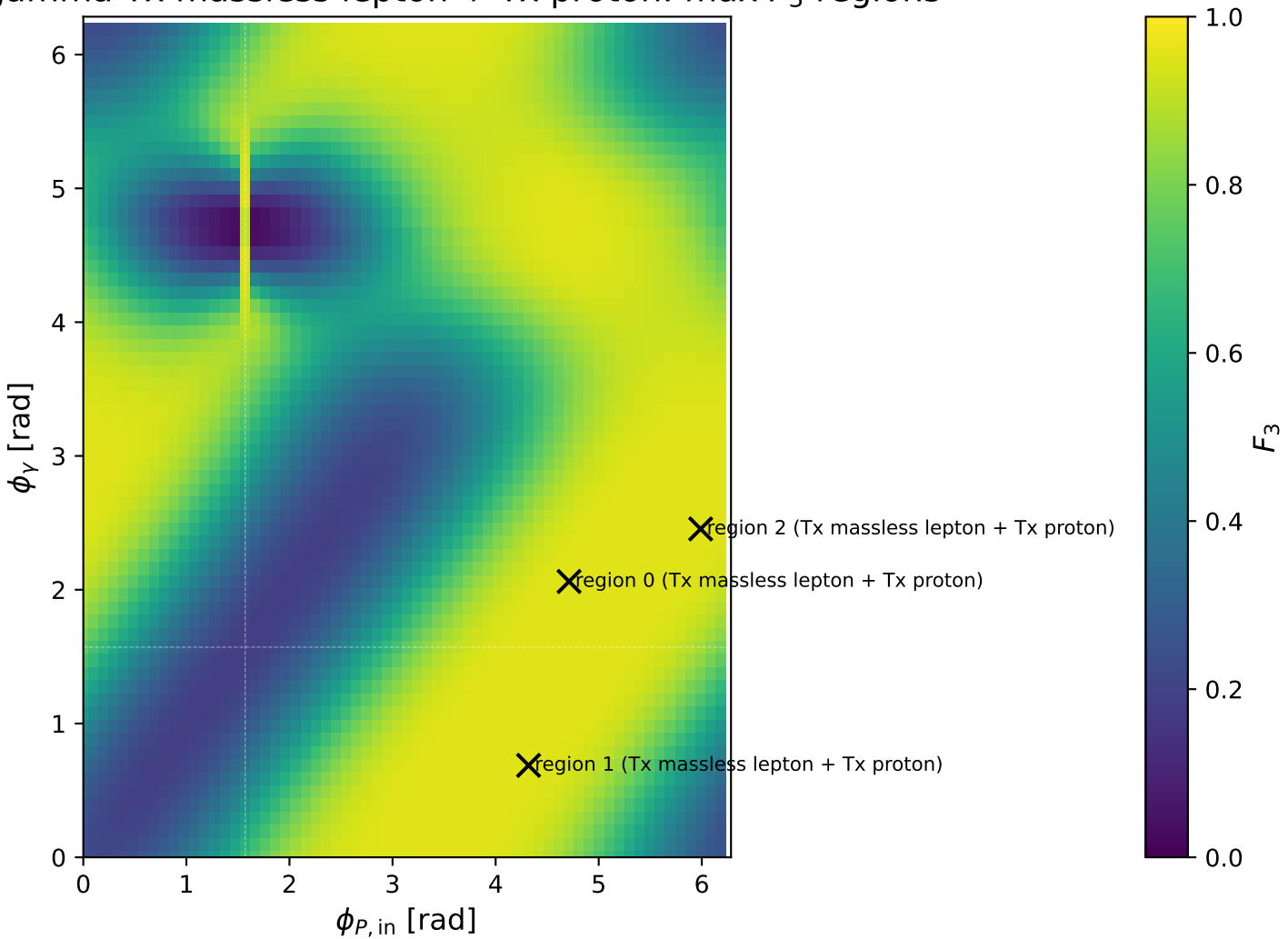
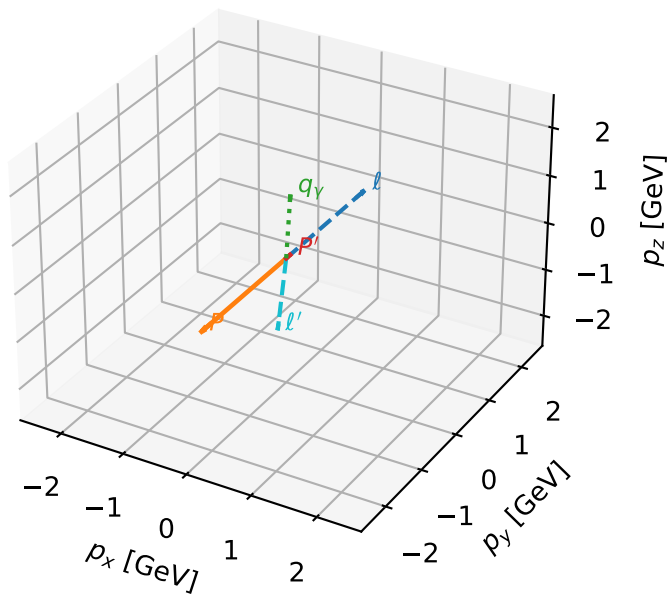


gamma Tx massless lepton + Tx proton: max F_3 regions



Tx massless lepton + Tx proton characteristic max F_3 configuration

3D momenta

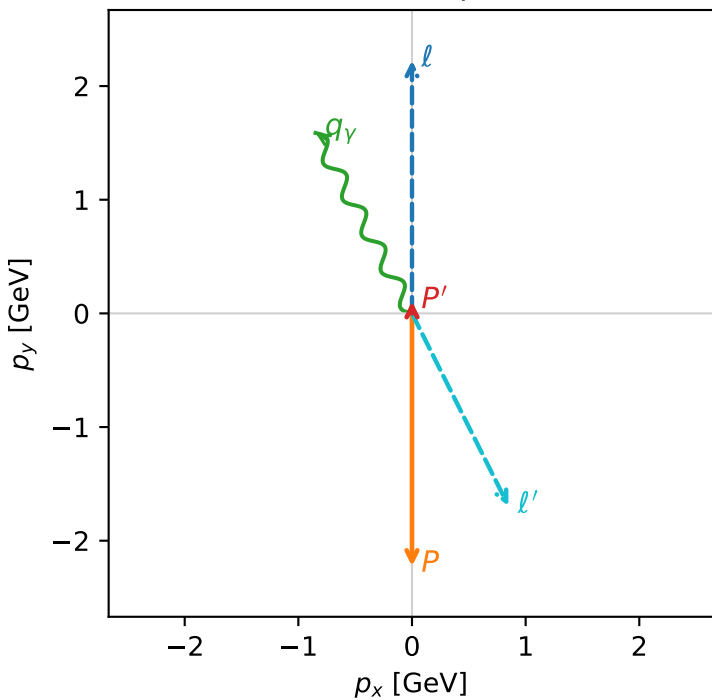


f3_high_egamma_txtx_region_0 $F_3=0.952541$
 region: high_Egamma_TxTx_region_0
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=0.106402$
 $\theta_{in}=1.57079$, $\phi_l=1.5708$, $\phi_p=4.71239$
 $E_\gamma=1.8$, $\phi_\gamma=2.06167$
 $Q^2=15.964$, $x_B=0.839123$, $t=-3.27157$, $W^2=3.94048$, $y=0.921917$
 $\theta(\ell', \gamma)=178.483$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 0.0000$ $p_y= 2.2244$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.0000$ $p_y= -2.2244$ $p_z= 0.0000$
 ℓ' $E= 1.8945$ $p_x= 0.8485$ $p_y= -1.6939$ $p_z= 0.0000$
 P' $E= 0.9440$ $p_x= 0.0000$ $p_y= 0.1064$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -0.8485$ $p_y= 1.5875$ $p_z= 0.0000$

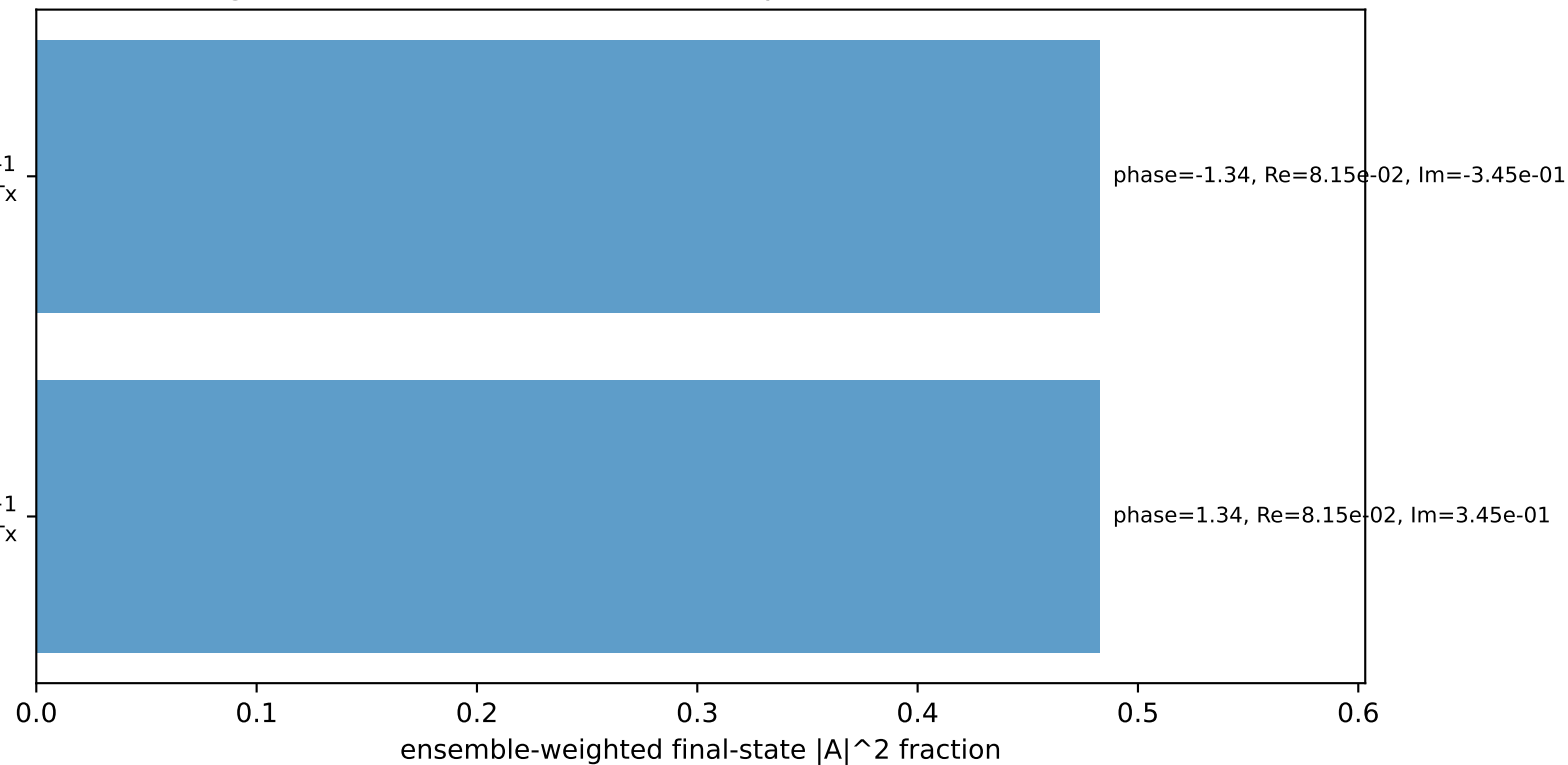
Transverse plane



$h_e=-1, h_p=-1, h_\gamma=-1$
 electron Tx, proton Tx

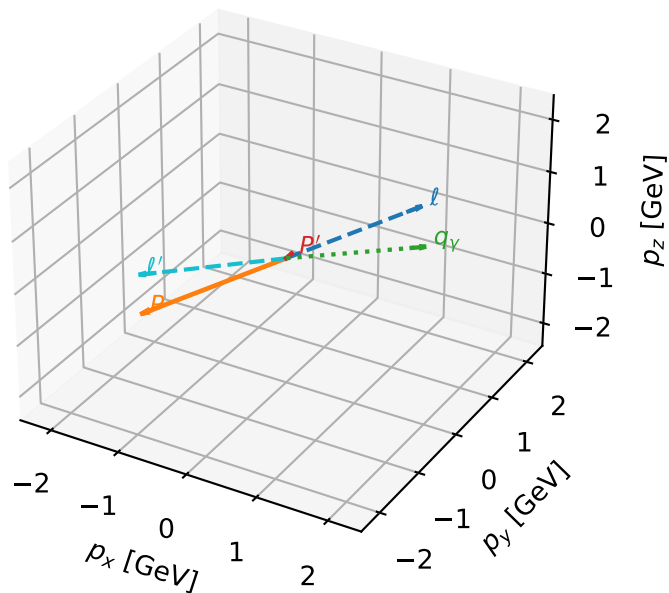
$h_e=+1, h_p=+1, h_\gamma=+1$
 electron Tx, proton Tx

Leading initial-ensemble/final-state components (N=2, retained=96.5%)



Tx massless lepton + Tx proton characteristic max F_3 configuration

3D momenta

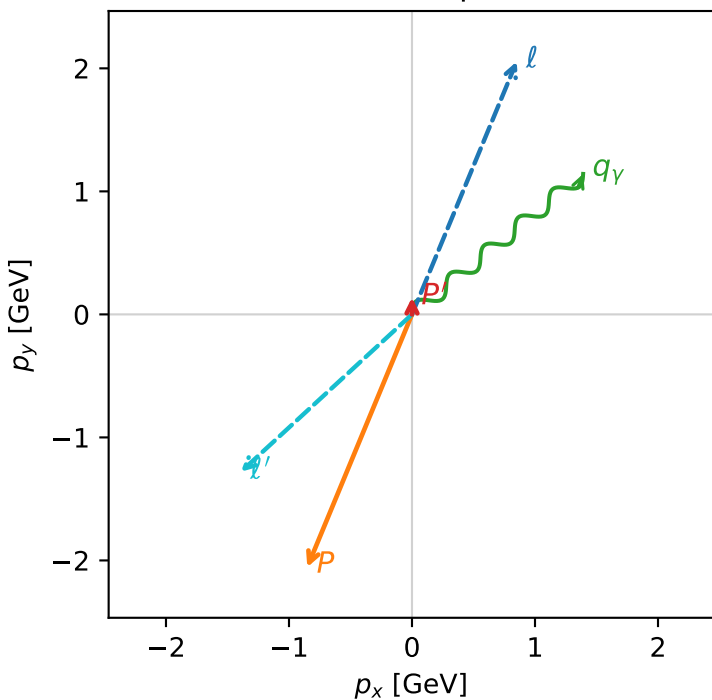


f3_high_egamma_txtx_region_1 $F_3=0.952428$
 region: high_Egamma_TxTx_region_1
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|T_{X,\ell_0}\rangle \otimes |T_{X,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=0.137832$
 $\theta_{in}=1.57079$, $\phi_\ell=1.1781$, $\phi_p=4.31969$
 $E_\gamma=1.8$, $\phi_\gamma=0.687223$
 $Q^2=16.0391$, $x_B=0.838104$, $t=-3.38431$, $W^2=3.97812$, $y=0.92738$
 $\theta(\ell', \gamma)=176.769$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= 0.8512$ $p_y= 2.0551$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= -0.8512$ $p_y= -2.0551$ $p_z= 0.0000$
 ℓ' $E= 1.8904$ $p_x= -1.3914$ $p_y= -1.2797$ $p_z= 0.0000$
 P' $E= 0.9481$ $p_x= 0.0000$ $p_y= 0.1378$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= 1.3914$ $p_y= 1.1419$ $p_z= 0.0000$

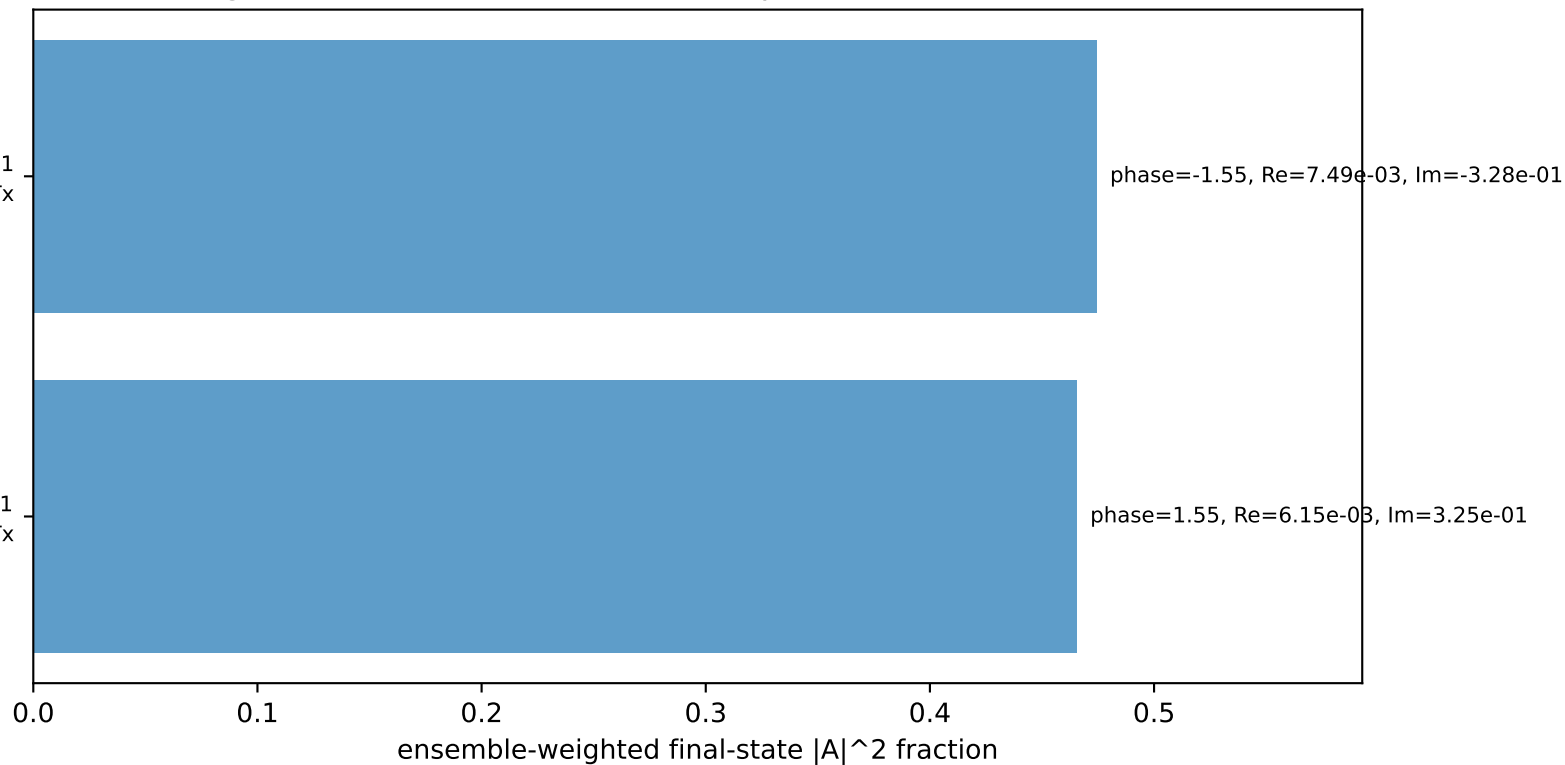
Transverse plane



$h_e=+1, h_p=+1, h_\gamma=+1$
 electron Tx, proton Tx

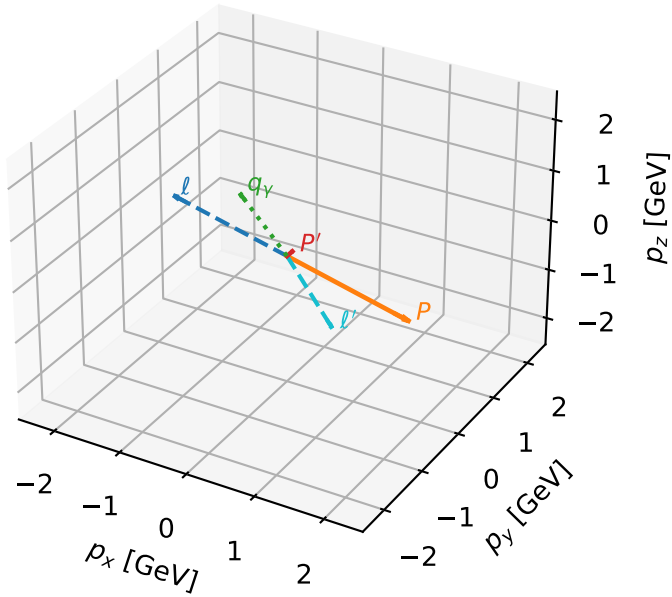
$h_e=-1, h_p=-1, h_\gamma=-1$
 electron Tx, proton Tx

Leading initial-ensemble/final-state components (N=2, retained=94.0%)



Tx massless lepton + Tx proton characteristic max F_3 configuration

3D momenta

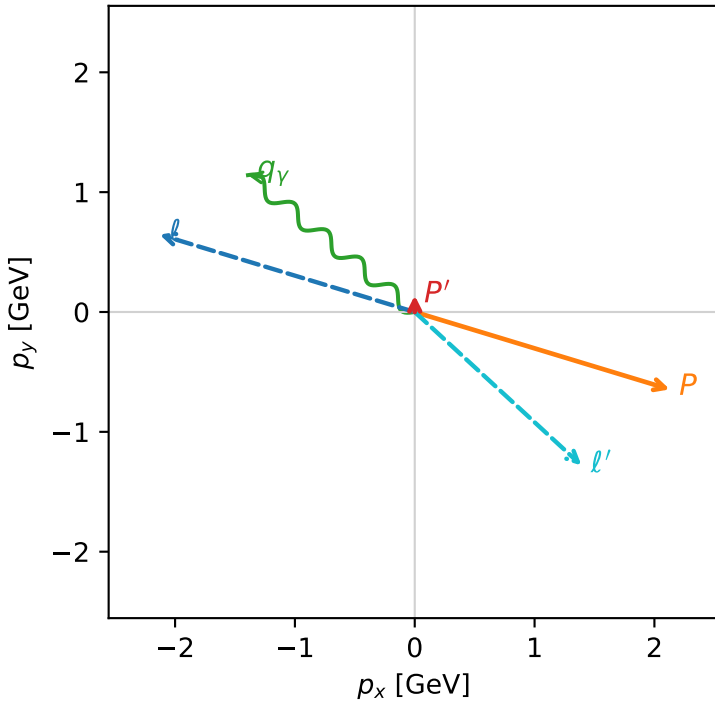


f3_high_egamma_txtx_region_2 F_3=0.952078
 region: high_Egamma_TxTx_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $|T_{x,\ell_0}\rangle \otimes |T_{x,p}\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22442$, $|\vec{P}'|=0.137832$
 $\theta_{in}=1.57079$, $\phi_\ell=2.84707$, $\phi_p=5.98866$
 $E_\gamma=1.8$, $\phi_\gamma=2.45437$
 $Q^2=15.9866$, $x_B=0.837658$, $t=-2.9958$, $W^2=3.97812$, $y=0.924835$
 $\theta(\ell', \gamma)=176.769$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2244$ $p_x= -2.1286$ $p_y= 0.6457$ $p_z= -0.0000$
 P $E= 2.4141$ $p_x= 2.1286$ $p_y= -0.6457$ $p_z= 0.0000$
 ℓ' $E= 1.8904$ $p_x= 1.3914$ $p_y= -1.2797$ $p_z= 0.0000$
 P' $E= 0.9481$ $p_x= 0.0000$ $p_y= 0.1378$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -1.3914$ $p_y= 1.1419$ $p_z= 0.0000$

Transverse plane



Leading initial-ensemble/final-state components (N=4, retained=96.5%)

